PROJECT PROPOSAL

Smart Agriculture: Using Machine Learning to Recommend Suitable Crops for Farmers

Category: Tabular Data | Multi-Class Classification

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Problem Statement

Smallholder farmers frequently lack access to agronomic expertise when deciding which crop to cultivate. Selecting an unsuitable crop given local soil and climate conditions leads to reduced yields, resource waste, and economic loss. This project addresses that gap by building a machine learning model that recommends the most appropriate crop based on measurable soil and environmental parameters. The problem is formulated as a 22-class classification task: given seven numerical features (Nitrogen, Phosphorus, Potassium, soil pH, temperature, humidity, and rainfall), the model predicts the optimal crop from a set of 22 species. Such a decision-support tool has direct practical value for extension services and precision-agriculture applications.

Challenges

Several challenges are anticipated. First, some crop classes may be under-represented in practice, so stratified splitting and class-aware metrics will be required to avoid misleadingly high aggregate accuracy. Second, certain soil and climate features are likely correlated (e.g., temperature and humidity), which can affect the stability of distance-based methods such as K-Nearest Neighbors; a correlation analysis will be conducted during EDA. Third, hyperparameter selection for Random Forest (number of estimators, maximum depth) and KNN (k, distance metric) must be performed with cross-validation rather than a single train/test split to avoid overfitting. Finally, interpreting and communicating results across 22 classes requires per-class precision, recall, and F1-score alongside a confusion matrix.

Dataset

We will use the publicly available Crop Recommendation Dataset from Kaggle (https://www.kaggle.com/datasets/atharvaingle/crop-recommendation-dataset). The dataset contains 2,200 samples with 100 observations per crop class, making it balanced across the 22 target categories. Each sample includes seven continuous numerical features: nitrogen (N), phosphorus (P), potassium (K), soil pH, temperature (°C), relative humidity (%), and rainfall (mm). The dataset will be downloaded from Kaggle and stored directly in the project repository.

Method and Algorithm

The project follows a supervised multi-class classification pipeline implemented in Python using scikit-learn. After exploratory data analysis and feature scaling (Standard Scaler), the data will be divided into training (80%) and test (20%) sets using stratified splitting to preserve class proportions. Three classifiers will be trained and compared: a Decision Tree (interpretable baseline), a Random Forest (ensemble method expected to reduce variance), and K-Nearest Neighbors (instance-based learner). Hyperparameter tuning will be carried out via k-fold cross-validation on the training set. The best-performing model will be selected based on validation macro-averaged F1-score and then evaluated on the held-out test set.

Evaluation

Model performance will be assessed using overall accuracy, macro-averaged precision, recall, and F1-score, and a 22×22 confusion matrix to visualize per-class errors. Because all 22 classes contain equal numbers of samples, overall accuracy is reliable; however, macro-averaged F1-score will be the primary selection criterion to ensure balanced performance across all crops. Cross-validation scores on the training set will be reported alongside test-set results to assess generalization. If time permits, a permutation feature-importance analysis will be conducted to identify which soil and climate variables contribute most to the predictions.